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Đề 2022

Câu 1: a) Ta có $\sum_{n=1}^{\infty} \frac{1}{2^n} \left(\frac{n+2}{n+1}\right)^{n^2}$

Xét $\sqrt[n]{\frac{1}{2^n}\left(\frac{n+2}{n+1}\right)^{n^2}} = \frac{1}{2}\left(\left(1+\frac{1}{n+1}\right)^{n+1}\right)^{\frac{n}{n+1}}$

$= \frac{1}{2} \cdot e > 1 \Rightarrow$ Chuỗi phân kì

b) Ta có $\sum_{n=1}^{\infty} \frac{(x+3)^{2n}}{n^2 4^n}$

Gọi $u = (x+3)^2 \Rightarrow \sum_{n=1}^{\infty} \frac{u^n}{n^2 4^n}$

Xét $\sqrt[n]{\frac{1}{n^2 4^n}} \to \frac{1}{4}$

Câu 2: $\lim_{x \to 0} \frac{\sqrt[3]{x+1} + \arctan x - 1}{\cos^2 x - \cos x + 2x}$

$\sim \frac{x^2}{\quad}$

Câu 3: $A = \int_1^{+\infty} \frac{dx}{x\sqrt{1+x^2}}$

$\lim_{x \to +\infty} \frac{x}{x\sqrt{1+x^2}} \sim \frac{1}{x^2}$ mà $\int_1^{+\infty} \frac{dx}{x^2}$ hội tụ

$\Rightarrow A$ hội tụ

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$$B = \int_0^1 \frac{\sqrt{x}}{e^{\sin x} - 1}\, dx$$

Xét $\lim\limits_{x \to 0} \frac{\sqrt{x}}{e^{\sin x} - 1} \sim \frac{1}{\sqrt{x}}$

Mà $\int_0^1 \frac{dx}{\sqrt{x}}$ hội tụ $\Rightarrow$ B hội tụ

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Đề 2023

Câu 1: a) Tacó $\sum_{n=1}^{\infty} \frac{1}{3^n}\left(\frac{n+2023}{n+2022}\right)^{n^2}$

* Xét $\sqrt[n]{\frac{1}{3^n}\left(\frac{n+2023}{n+2022}\right)^{n^2}} = \frac{1}{3}\left(\frac{n+2023}{n+2022}\right)^{n}$

$= \frac{1}{3}\left(1+\frac{1}{n+2022}\right)^{n}$

$= \frac{1}{3}\left[\left(1+\frac{1}{n+2022}\right)^{n+2022}\right]^{\frac{n}{n+2022}} = \frac{e}{3} < 1$

⇒ Chuỗi hội tụ

b) Ta có $\sum_{n=1}^{+\infty} \frac{(x-1)^n}{2^{n+1}\sqrt{n^2+\ln n}-\cos n}$

Câu 2: Xét $\lim_{(x,y)\to(0,0)} \frac{\cos 2y - \cos 2x}{2(y^2+x^2)}$

$TH_1: y=0 \Rightarrow \frac{\cos 2y - \cos 2x}{2(y^2+x^2)} = \frac{1-\cos^2 x}{2x^2} = \frac{2\sin^2 x}{2x^2}$

$= \left(\frac{\sin x}{x}\right)^2 = 1$ khi $x \to 0$

$TH_2: x=0 \Rightarrow \frac{\cos 2x - \cos 2y}{2(y^2+x^2)} = \frac{-2\sin^2 y}{2y^2} = -\left(\frac{\sin y}{y}\right)^2$

$= -1$ khi $y \to 0$

⇒ giới hạn phiếm tồn tại

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Câu 3: $A = \int_1^{+\infty} \frac{(2x^2+x+1)dx}{e^{4x}+x^3+3}$

Xét $\lim_{x\to+\infty} \frac{2x^2+x+1}{e^{4x}+x^3+3} \sim \frac{2x^2}{e^{4x}}$

Lấy $\frac{1}{x^2}$ để so sánh: $\lim_{x\to+\infty} \frac{x^2(2x^2+x+1)}{e^{4x}+x^3+3} = 0$

Mà $\int_1^{+\infty} \frac{dx}{x^2}$ hội tụ $\to$ A hội tụ

$B = \int_0^1 \frac{2x+1}{(x^3-1)(3x+4)(5x+7)}$

Xét $\lim_{x\to 1} \frac{2x+1}{(x^3-1)(3x+4)(5x+7)} = \frac{1}{x-1} \cdot \frac{2x+1}{(x^2+x+1)(3x+4)(5x+7)}$

$\sim \frac{1}{81} \cdot \frac{1}{x-1}$

Vì $\int_0^1 \frac{dx}{1-x}$ phân kì $\to$ B phân kì

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~~Câu 1:~~

Đề 2024

Câu 1a) Xét hội tụ của chuỗi $\sum_{n=1}^{\infty} \frac{n!\,3^n}{n^n}$

Ta có $\frac{a_{n+1}}{a_n} = \frac{(n+1)!\cdot 3^{n+1}}{(n+1)^{n+1}} \cdot \frac{n^n}{n!\,3^n} = \frac{(n+1)\cdot 3\cdot n^n}{(n+1)^n \cdot (n+1)}$

$= \frac{3\cdot n^n}{(n+1)^n} = 3\frac{3}{\left(1+\frac{1}{n}\right)^n}$

Mà khi $n\to\infty$: $\left(1+\frac{1}{n}\right)^n \to e$

$\Rightarrow \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \frac{3}{e} > 1 \Rightarrow$ chuỗi phân kì

b)

Câu 2: Ta có $\lim_{(x,y)\to(0,0)} \frac{x\sin y}{xy + y^2}$

Đặt $y = kx \to$ ta có $\lim_{(x,kx)\to(0,0)} \frac{x\sin(kx)}{kx(x+kx)} \; kx^2 + k^2x^2 = \frac{\sin(kx)}{kx(1+k)}$

Xét TH: $k = 1 \Rightarrow \lim_{(x;kx)\to(0,0)} \frac{\sin x}{x\cdot 2x} = \frac{1}{2}$ (do $\frac{\sin x}{x}$ tiến đến 1)

Xét TH: $k = 2 \to \lim_{(x,2x)\to(0,0)} \frac{\sin(2x)}{6x} = \frac{1}{3}\cdot\frac{\sin 2x}{2x} = \frac{1}{3}$

$\Rightarrow \lim_{(x,2x)\to(0,0)} \frac{\sin(kx)}{kx^2 + k^2x^2}$ phụ thuộc $k \Rightarrow$ giới hạn ko tồn tại

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Câu 3: $A = \int_1^{+\infty} \frac{(3x+1)dx}{x^2+x-2} = \int_1^2 \frac{(3x+1)dx}{x^2+x-2} + \int_2^{+\infty} \frac{(3x+1)dx}{x^2+x-2}$

Xét $\lim_{x \to 1^+} \frac{3x+1}{x^2+x-2}$;

$= \lim_{x \to 1^+} \frac{3x+1}{(x-1)(x+2)}$

$= \lim_{x \to 1^+} \frac{\frac{11}{3}}{x-1} + \frac{\frac{5}{3}}{x+2}$

$\sim \lim_{x \to 1^+} \frac{11}{3} \cdot \frac{1}{x-1}$

Vì $\int_1^2 \frac{dx}{x-1}$ phân kì nên ~~lim~~ $\int_1^2 \frac{(3x+1)dx}{x^2+x-2}$ phân kì

$\Rightarrow$ A phân kì

B: $\int_1^{+\infty} \ln(\cos\frac{1}{x})dx$

Xét $\lim_{x \to +\infty} \ln(\cos\frac{1}{x}) \sim \lim_{x \to +\infty} \frac{-1}{2x^2}$

Vì $\int_1^{+\infty} \frac{-dx}{2x^2}$ hội tụ $\rightarrow$ B hội tụ

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Đề 2025

Câu 1: a) Xét $\sum_{n=1}^{\infty} \left(\cos\frac{1}{n}\right)^{-4n^3+n^2}$

Xét $\sqrt[n]{\left(\cos\frac{1}{n}\right)^{-4n^3+n^2}} = \left(\cos\frac{1}{n}\right)^{-4n^2+n} = e^{(-4n^2+n)\ln\cos\frac{1}{n}}$

Xét $(-4n^2+n)\ln\cos\frac{1}{n}$ khi $n \to \infty$

$\Rightarrow (-4n^2+n)\ln\cos\frac{1}{n} \sim (-4n^2)\ln\left(\cos\frac{1}{n}-1+1\right)$

$\sim (-4n^2)\ln\left(-\frac{1}{2n^2}+1\right) \sim (-4n^2)\left(\frac{-1}{2n^2}\right) \sim \frac{4n^2}{2n^2} = 2$

$\Rightarrow \sqrt[n]{\left(\cos\frac{1}{n}\right)^{-4n^3+n^2}} = e^2 > 1 \Rightarrow$ chuỗi phân kì

b) Xét $\sum_{n=1}^{+\infty} \frac{(2x+3)^n}{3^n} = \sum_{n=1}^{+\infty} \left(\frac{2x+3}{3}\right)^n$

$\Rightarrow$ chuỗi hội tụ khi $\left|\frac{2x+3}{3}\right| < 1 \Rightarrow -3 < 2x+3 < 3$

$\Rightarrow -3 < x < 0$

$\Rightarrow$ Miền hội tụ tại $x \in (-3;0)$

Câu 2: Xét $\lim_{(x,y)\to(-1;0)} \frac{(x+1)^2 y^3}{(x+1)^6 + y^9}$

Đặt $u = x+1 \Rightarrow \lim_{(u,y)\to(0,0)} \frac{u^2 y^3}{u^6+y^9}$

Xét ~~$y = Ku \Rightarrow$~~ TH $y = Ku$: $\lim_{(u,Ku)\to(0,0)} \frac{u^2 K^3 u^3}{u^6 + K^9 u^9}$

$= \frac{K^3 u^5}{u^6(1+K^9 u^3)} = \frac{K^3}{u(1+K^9 u^3)}$

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$\Rightarrow$ Khi $K=0 \Rightarrow \lim\limits_{(u;Ku)\to(0,0)} \dfrac{K^3}{u(1+K^9u^9)} = 0$

Khi ~~$K\neq 0$~~ $K=1 \Rightarrow \lim\limits_{(u,Ku)\to(0,0)} \dfrac{K^3}{u(1+K^9u^2)} = \infty$

$\Rightarrow$ ~~Hay~~ $\lim\limits_{(u,Ku)\to(0,0)} \dfrac{K^2}{u(1+K^9u^2)}$ phụ thuộc K $\Rightarrow$ giới hạn không tồn tại

Câu 9: $A = \int_0^1 \dfrac{\sin(1-x)\,dx}{\sqrt[3]{1-x^5}\,(e^{1-x^2}-1)}$

Xét $\lim\limits_{x\to 1^-} \dfrac{\sin(1-x)}{\sqrt[3]{1-x^2}\,(e^{1-x^2}-1)}$ ~

$\sim \dfrac{1-x}{\sqrt[3]{(1-x)(1+x+x^2)}\,(1-x^2)}$

$\sim \dfrac{1-x}{\sqrt[3]{3}\,(1-x)^{\frac{1}{3}}\cdot 2(1-x)}$ ~

$= \dfrac{1}{2\sqrt[3]{3}} \cdot \dfrac{1}{(1-x)^{\frac{1}{3}}}$

Mà $\int_0^1 \dfrac{dx}{(1-x)^{\frac{1}{3}}}$ hội tụ $\Rightarrow$ A hội tụ

$b = \int_0^{\infty} \dfrac{(x+2)\,dx}{x^2(x+3\cos x+5)}$

$= \int_0^1 \dfrac{(x+2)\,dx}{x^2(x+3\cos x+5)} + \int_1^{\infty} \dfrac{(x+2)\,dx}{x^2(x+3\cos x+5)}$

Mà $\int_0^1 \dfrac{(x+2)\,dx}{x^2(x+3\cos x+5)}$ phân kì do mũ ở mẫu $= *3?$

$\Rightarrow$ B phân kì

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